

Motor Engineering Technical Data Manual

1. Three-Phase Induction Motor Full Load Current Ratings (IEC 60034-1)

Motor Power	230V 3Ph	400V 3Ph	415V 3Ph	480V 3Ph	690V 3Ph	Eff %	Cos Phi
0.37 kW	1.85 A	1.05 A	1.01 A	0.90 A	0.60 A	66.0%	0.72
0.75 kW	3.25 A	1.85 A	1.78 A	1.60 A	1.07 A	73.0%	0.76
1.50 kW	6.00 A	3.45 A	3.30 A	2.95 A	2.00 A	78.5%	0.79
2.20 kW	8.50 A	4.90 A	4.70 A	4.20 A	2.83 A	81.0%	0.81
4.00 kW	14.80 A	8.50 A	8.15 A	7.20 A	4.91 A	84.5%	0.83
7.50 kW	26.50 A	15.20 A	14.60 A	13.00 A	8.78 A	87.5%	0.85
11.0 kW	38.00 A	22.00 A	21.00 A	18.80 A	12.70 A	89.0%	0.86
15.0 kW	51.00 A	29.50 A	28.20 A	25.20 A	17.00 A	90.0%	0.86
22.0 kW	73.50 A	42.50 A	40.50 A	36.20 A	24.50 A	91.5%	0.87
30.0 kW	99.00 A	57.00 A	54.80 A	48.80 A	32.90 A	92.5%	0.88
45.0 kW	146.00 A	84.50 A	81.00 A	72.00 A	48.80 A	93.4%	0.88
75.0 kW	239.00 A	138.00 A	132.00 A	118.00 A	79.70 A	94.2%	0.89
110 kW	348.00 A	201.00 A	192.00 A	171.00 A	116.00 A	95.0%	0.90

2. Motor Terminal Block Connections (Star & Delta Bridging)

Terminal layout: W2-U2-V2 (top row) and U1-V1-W1 (bottom row)

Star (Y) Connection: Single horizontal link bar shorting W2-U2-V2.

- Supply connected to U1, V1, W1. $V_{winding} = V_{line} / \sqrt{3}$.

Delta (D) Connection: Three vertical link bars (W2-U1, U2-V1, V2-W1).

- Supply connected to U1, V1, W1. $V_{winding} = V_{line}$.

3. Power Factor Correction & Capacitor Sizing

$Q_c = (P_{shaft} / \text{efficiency}) * [\tan(\text{acos}(PF_1)) - \tan(\text{acos}(PF_2))]$

Delta bank capacitance: $C = (Q_c * 10^9) / (3 * 2 * \pi * f * V^2)$ [microfarad]

Important Relay Rule: When capacitors are placed at motor terminals, the thermal overload relay must be reset to the reduced line current:

$I_{relay} = I_{original} * (PF_{original} / PF_{corrected})$